

Building Agents on AWS

The AWS agent toolkit · Strands Agents · Bedrock AgentCore · Keeping agents in bounds
· Amazon Q and Amazon Quick

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The AWS agent toolkit

If you want to...	Use
Write your own agent in code	Strands Agents
Coordinate several agents	Agent Squad
Run agents in production	Bedrock AgentCore
Buy rather than build	Amazon Q Business
Give the business a self-serve tool	Amazon Quick



You'll still see Bedrock Agents in older tutorials.

AWS is retiring it — new work goes to Strands and AgentCore.

Strands Agents

Write the agent in code. The model works out the steps.

You give it three things:

-  A goal — what you want done
-  A list of tools it may use
-  A model to do the thinking

 **Older frameworks made you draw the whole flowchart. Here you describe the tools well and let the model plan.**

Your job isn't easier — it's different. It moves from writing logic to choosing which tools the agent may touch.

Amazon Bedrock AgentCore

Writing an agent is the easy part. Running one safely is the work.

Piece	What it handles
Memory	Remembering across conversations
Gateway	Turning your existing APIs into tools
Identity	Who the agent acts as, and what it may touch
Runtime	Actually running it, at scale



Identity is the one to notice.

The agent should act as the user, not as an admin — so if that person can't see the payroll file, neither can the agent working for them.

Keeping agents in bounds

Policies	Evaluators
Rules about what it MAY do	Tests that check it DID it well
"Never refund over £500 unapproved"	"Did it pick the right tool?"
Checked while it runs	Run before you ship a change



Why this matters more for agents than chatbots:

A chatbot with a bad answer is embarrassing. An agent that gets it wrong has done something. The mistake isn't a sentence — it's an event.

Amazon Q Business

A ready-made assistant that connects to what you already use.

-  Connectors for SharePoint, Confluence, Salesforce, S3 and more
-  Respects your existing permissions — people only see what they already could
-  Q Apps — describe a small tool in plain English and get one



Example

"Summarise this quarter's customer complaints by theme." Q Business reads the ticket system and answers — no build required.



AWS is retiring Q Business — check what's current before building on it.

Amazon Quick

The newer, self-serve version — built for people who don't write code.

Piece	What it does
Spaces	Where a team's knowledge lives
Connectors & Index	Pulls your data in and makes it searchable
Flows / Automate	Turns a repeated task into a click
Research & Apps	Produces an actual deliverable



Same lesson as BLA-01: it's only as good as the data you ground it in. Point it at a messy shared drive and you get messy answers.

Key takeaways

- 1 Strands to write it · AgentCore to run it
- 2 Skip Bedrock Agents — AWS is retiring it
- 3 Identity matters: the agent acts as the user
- 4 Policies while it runs · Evaluators before you ship
- 5 Q Business / Quick if you'd rather buy than build

Next · Part 3: the operational efficiency and optimization.